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Well, yeah, no. Essentially, imagine if two AIs were playing chess instead of chess, instead of a chess board, an S3, a rendered S3, an R4, Riemann 4 topology.

that essentially represents the universe was the board so imagine the universe is the board and mathematical objects anything that couldn't be generated geometrically with mathematics are the chess pieces so imagine if you're playing chess and you wanted to make a certain move with a certain card or i mean certain piece that you didn't have imagine if instead of being like oh man i should get my pawn to the end so i can turn it into a queen's

so I can use the queen, do X amount of moves. Imagine if you could just generate it and just generate it. And you generate any sort of geometric object that everything is bounded to physics, right? So the way that a cube has different physical properties than a sphere, not just in terms of geometry, but in terms of dynamics, like fluid dynamics, aerodynamics, this and that.

right? It's a whole, it's a multitude of different possible transformations, I guess. So imagine if, and now keep in mind, all of this stuff is being represented for the AI is being represented mathematically. It's not generating structures that it's calling a star or a sun or a black hole. It's just generating mathematical structures that it's pulling from physics libraries and

we also have extensively formulated uh physical theories into math lib and lean etc for the sake of just architecture it's using physics to it's making a structure that you know we you could consider a black hole but in this case for the agent it's it's more so just

a geometric object in this simulated this s3 space that has the properties now everything that's generated once again imagine just like anything in this universe there's an energy there's mass there's it's almost fractal right like the universe you could posit has some sort of finite

boundedness or some sort of a total amount of energy total amount of mass total amount of This and that is not this vet and everything inside the universe has a total amount of this or a total amount of XYZ and this now and it's the same right and planets have mass and then our planets have all these physical properties that on Earth everything has this physical properties The only difference is right the scale. So, quanta, you know, light has mass and speed, or light is masses, but light has speed versus a car. Cars have speed, planets, their planetary motions have speed. The difference is at that level, that quantum scale, right? The speed is exponentially faster than any material object bounded to gravity or weighted by gravity in that sense.

And this game does simulate weak force, gravity force, all this shit. But honestly, physics is more so just... It's mathematical physics would be an easier way to put it. Mathematical physics. And now, so imagine if you're... In this case, these agents, they're playing a game where a universe has been instantiated. And...

The goal of one is to perpetuate it, you know, to have the charge accumulation not be zero. The other is to hopefully bring it to zero. So say X is, you know, however many computations deep into the game, this game has all the particles and all the atoms and

whatever possible to make essentially a sun, a mathematical object that represents the sun. You know, take, and obviously the atoms would be represented in their own mathematical properties based on real physics. And so imagine one makes a sun and now the adversarial one is like, okay, I'm going to make a black hole right next to the sun to cancel it out. Now,

the physics of this reality wouldn't permit that, right? Nothing's just, you can't just spawn a black hole from nowhere. There needs to be some sort of, you know, collapse of a star or amount of whatever the fuck to permit its existence. So this agent, if it did want to make a black hole, would have to make all the things that will lead to whatever will be a black hole and hope that this other adversary, hope it does, hope it has a substantial effect.

So it's a game in a semantic sense, it's a simulation in a semantic sense, whatever. A mathematical universe in a, actually pretty much a realist sense, but it's essentially like we live in a universe where we're discovering, we live in an already finished universe, existing universe, and we're essentially discovering all the rules to it, you know, through math and physics. So imagine if you took all the rules that we have discovered in math and physics, and then you use those rules

as the ruleset for a game. And in this game, those rules and everything that they permit permit a lot of other things that we might not even have accounted for in a strictly mathematical, computational universe, you know? So the aim isn't really to replicate the universe. That's obviously...

Beyond the scope or not just this game but of physics and math itself the goal is more so just to have a mathematical universe in which the Transformations the abstractions and this and that are one to one with our reality. So in For example, what we'd want if one plus one equals two in real life We want one plus one equal to in this simulation and whatever it does with

Whatever it could be one and whatever it could be to whatever whatever it does with those symbols those variables those numbers Could also you know have an effect in real life the same way that somebody could some the idea of dark matter. Uh Was came up with by I forget like plank or the rock maybe like they came up with it before they actually discovered it and you can do this stuff using mathematics and

because it's not simply just some made up arbitrary symbol, but it's as close to reality as abstraction gets. So this is the aim of this project is to do something that accomplishes that. And it's not, this is why I say it's a game in a sense that there's a sort of a zero sum competition. You know, one agent is trying to bring something towards zero or keep it as asymptotic to zero as possible to,

The other is trying to expand the mathematical universe and create more geometric objects that are way more resistant to symmetry breaking and decay and this and that. And we might also discover...

that these strategies, these applications are so mathematically sound and proven within this game by these agents that we could actually take it, apply it to real life and solve genuine problems as well.

But that's a bit of a cherry on top. Regardless of if that is the case, at the very least, this will still be an adversarial mathematical universe in which

It may only give pure mathematics or abstract mathematics solutions that maybe, you know,

Just like the Navier-Stokes, for example, right? You can use that equation, but at a certain point you get blow up and near infinite velocities, which is not possible in our universe because of molecular or because molecules things become, you know, fluids break down into molecules before they have the chance to be whatever the fuck is capable of having an infinite velocity, whatever sort of minutiae.

And the game might do something like that where it's able to use mathematics bound into laws of physics in such a way that, you know, there's it's a universe of its own. And that's the goal of this. It's like imagine if we were to have another universe that is abiding by the same mathematics, but a completely different ontology, at least from.

In this case, ontology is two agents adversarial with an aim that reflects biology or that reflects biological systems. But imagine if biological systems were using abstract mathematics

To get it to achieve its aim so imagine if instead of oh I want to you want to move a car from point A to point B the human thing to do would be move what's the what's the fastest way well, how can we get a straight based on a straight line from A to B how can we stay as close to that as possible and Get from point A to point B. They say I might however decide, you know Maybe make a manifold structure that after say

you know x amount of distance over time if something were to traverse the structure it would make its way from point a to point z and then from point z to point b and or however many computations in between so it's an ai that's using mathematics but it's not doing it to solve a human problem there's it's not anthropocentric it's a game made by man sure but

That's it's almost like a bit of a Frankenstein situation of Pandora's box high situation where it's like once it's, you know, once it's made, then from there, it's in a sense, as a mind, a life of its own. And from there, it just becomes whatever anybody wants to make of it. And to touch on something, somebody asked me, OK, this is such a.

a genius idea. Why is it not being done? There has to be a catch. It's so genius, but open AI or anthropic or any of these AI or deep mind itself, right, they're not doing this. And while I'd say, okay, there's 10 billion people on earth, 99.9% are inherently, you know, objectively just not focused on this problem of

simulation, computation, abstract mathematics, training AGI or training AI in hopes of AGI, this and that. And of the ones that are doing it, everybody, this is it. I guess I caught lightning in a bottle with the idea, you know, but...

I'd say the only reason it hasn't been thought of yet is it hasn't been thought of yet. That's like, oh, there's everybody on the world knows what a phone is. So how could you be the one that conjured up the iPhone and the screenless phone? You know, if you ask that to Steve Jobs, what's the reason? By so many things that have nothing to do with you. And if you happen to just get an idea like that, you know, you luck out.

But it's not something to be arrogant about. I'm not claiming that I've come up with this. This is built on the past 10,000 years of math. And theory, and of course, there's all these sources that are cited as well. It's not something that exists in a vacuum. I just happen. It's like something you made the recipe, nature made the ingredients, and some people did a million things. Now I'm just putting it together in such a way that

It's just a matter of imagination. I like games. I like math. I like code, et cetera, et cetera. So put it all into one. When I say game of games, it's more of a game, a universe out of game, you know, a universe out of machine, a sort of universo ex machina ludicrous or some shit. And honestly, the bare minimum implications of this is that it's going to produce insight

it's almost a manhattan project type of thing where okay maybe it won't revolutionize shit but it's definitely gonna you know it's not like the manhattan project solved fusion reactors but it for sure um for sure brought nuclear fusion or fission whatever the fuck into mainstream thought and research and shit

you know so whatever it actually does do who knows but the whole point is to make it and see you know otherwise it's it's not this is it's just like any conjecture make the conjecture and then see it's proven over time but yeah i guess you could consider a research

It's really I'm not even concerned with the semantics of it. I like who cares because I thought what you can call it who you just could be called as the universe the universe or called reality reality doesn't make it whatever if we're calling it so There's no it does just because it doesn't have a box for us to put it in You know, they'll probably be a word made for something like this a sort of

game universe simulation, but I would push back on the notion that this is a universe physics simulator and it's a physics simulator but Not necessarily our universe that is being simulated It is whatever can be simulated given the possibility of being able to generate mathematical structures in an s3 in a Riemann for topology, so

whatever that means, you know, it's beyond my scope as well. I'm not a mathematician the same way that Walt Disney was in architecture, you know, but he still made fucking Disneyland. And what I'm doing is just that. For me, it's speculation and prompt engineering, but it's the AI that has the logic and it's the lean programming language and math and physics that are grounding it in reality, quite literally.

So my goal, my job essentially is to be the delusional thinker and imagine this idea to its limits, exhaust its potential and make sure that AI and other researchers and the people building this are able to do that.

But the one thing that will not be compromised is this game, because if it's not makeable for us, at least we're going to theorize it in such a way that it will be makeable once humans have whatever is necessary. But I don't even believe that would be the case. I believe that with what's already existing now, math, physics, lean, libraries, ingenuity and creative,

you know, construction, et cetera, something that accomplishes the goals of this will be possible, even if it's not one to one with how it's being positive, positive right now. So, but really it is. Imagine, you know, if, if,

I'd say in terms of why does it produce things that a human would understand? Well, as a human, say, I want to make I want to turn one to zero. This is the first thing I would do. In fact, in a lot of cases, the only thing I would think to do is I want to make one zero. I'd have to just subtract, you know, combine it with negative one, add negative one, and then it cancels out.

In the sense of the game that's being played with these agents, imagine if instead of, oh, how can I get $1 + 1 = 1$, or how can I get $1 = 0$? $1 + (-1) = 0$? Nah. How about I actually multiply 1 by the exponent of 148 and then divide that by this and then add that, et cetera, et cetera, and then get 0.

And another interesting thing is it's solving mathematical problems using physics. So it's solving problems that exist in the abstract sense of math and physics, but they may not exist in our universe or yet. But theoretically, if there is infinite worlds, there is a universe. Infinite worlds that follow the same underlying mathematics and physics.

Then shit, this AI is simulating one of them. It's just not an anthropocentric one. And it might not even be one that humans understand. We can understand the mathematics, hopefully, but besides that, who knows? And the reason that this game creates such a general intelligence is because it is learning, simply learning to solve any possible problem.

mathematical or physical problem without an objective that has to do with anything in reality for humans. So we might think, how can we use calculus to get a rocket to the moon? These agents are thinking, how can we use calculus to create a transformation in this universe that we've made for mathematics?

to simulate this physical theorem that says, postulates, if we create an object with these properties, it will negate an object or the effects of an object with other properties. So if you were to make it anthropocentric, it would have to be from the perspective of humans who have reached such a synthesis with mathematical conjecture and theory that they approach reality itself as a

a mathematical structure. They don't see the earth as earth or, you know, people as people to see everything at their base, you know, the machine language of reality essentially, or the abstract mathematics behind the machine language of reality, not the semantic language of reality. So yeah, it's really interesting stuff. And this is why it needs to be done, you know, without compromise, because

It's like imagine if people decided not to make neural networks because the framework was too complicated to conceive now we have AI and something like Claude grog Dut whatever that this is fully doable. The only thing that could stop it is if certain problems are seen as permanent obstacles versus things to work around and

And this is why it takes ingenuity along with discipline. And because this AI may hallucinate stuff, it will very, very much be discouraged and rigorously refined if it does. But this is not the making of a game that just has to be good in theory or a game that can just be played. It is, I'll simply say a structure, you know, I'll call this a structure, right?

a systemic structure for research in abstract mathematics, physics, solution, anything. It's like trying to describe light to someone blind. This is a whole new arena. And it's honestly, somebody else told me, they're not surprised that I came up with it rather than Claude or Chaudhry or any of these companies.

they're more so surprised that it's taken this long for it to come up with and it also highlights the bottleneck that intelligence faces in the modern world which is it's almost animal husbandry like of take the smart people round them up put them in these niches or these areas and have them study for these problems and get the money from research for these methods and

And it makes it so obvious why it took so long, because these are abstract. It's abstract mathematics and physics at the core of this. But the solution is not something that computer science or abstract mathematicians or physicists would have come up with because they're too deep into that bottleneck to where there needs to be somebody with a childish like.

hope optimism that you know a level of Conjuring this idea of something that can exist Because they don't even have a notion of can't you know? And that's part of why we were able to get around the young mills gap or how we decided to add the novers Nover charge accumulation as a metric to keep it working and

It's because a regular mathematician or physicist or someone would have stopped there and decided, oh, look, it's not real in reality. So we can't do this. This is a problem that's not even solved in reality. So we can't solve it here. And if you were to tell that to a kid, they would simply say, why? Find a way. You know, there is a way. And it's not constrained to mathematics. It's constrained to logic.

I mean, it's not constrained to human imagination. So even when these LLMs or these AIs say, oh, this is a problem, it's because it's trained off of human data sets, human semantics, and it still thinks like that. But this project requires a version of thinking that is almost as alien as these agents and one that does not even view any sort of problem as deterrent.

It's actually encouraged because the more problems we find, that highlights that we're looking in the real world. It highlights that we're battle testing this against reality and proofs. Because if we were to conceive this idea and we couldn't find any problems of it, then that means it's in la-la land. It's so out there that it's science fiction, not science, therefore pseudoscience is okay.

and that is the last thing that we want. There should be no sort of pseudo mathematics, pseudo code, pseudo science, pseudo theory. The only use of hard science speculation is to look at the technical specs of this project, look at the ambitions, and because it doesn't exist, because there isn't current literature on it, we have to just

speculate. But it's not speculation based in fantasy. It's speculation based in technical specs and architecture, which is another clarification. And as to why is it done in that way? Why not, right? We could give only dissertations, only technical specifications, or we could also make it a bit interesting and add in some pure speculative

non-fictional fiction I guess, you know, and it's all for the sake of abstraction and you know somebody brought it up in one of the debates that perhaps that's the truth that might be discovered from this game that there actually isn't a difference between truth and abstraction and abstraction contains truth or it might be synonymous with it. These are all things that we are going to find out and it's very exciting stuff that

We've also been told by experts that this is very doable and it's actually codable. This is what's funny is the lean, the programming, the coding for this is not what's holding it back. We can do that easily. And it can be done even with something like Grok. This project is very doable. It's just maintaining rigor, integrity.

and getting it one-to-one, not with reality in the universe, but one-to-one with the physics and mathematics of the universe. Now, of course, that comes at the fundamental inherent problem. We are still discovering...

the mathematics and physics of this universe. Therefore, it is not fully comprehensive. And that is why the workaround was to, okay, fuck it, don't make it the exact, don't make the universe. You know, I'll repeat, but if there's a business or a company or researchers who want to take their ambitions in that direction,

of universal physics simulations and they want to take it in the direction of actually trying to create a one-to-one reality then they can do that for us we're taking it a step that it's almost like symbolic ai versus what we have uh we've generated this we're not trying to everybody and which is why another answer i said to someone why why is in deep mind and proper doing this was such a good idea

everybody is still thinking too anthropocentrically who gives a fuck about replicating the universe from the human perspective we are replicating or we attempt to be to generate have agents that can generate ones from mathematics from the underlying mathematical structure of the universe removed with all the ontology that humans have conceived the planets and stars and this and systems and this and that no no no this is more of like imagine if you just

stripped everything down to sort of mathematical, realist, object-oriented ontology of everything, you know? So it's taking simulation theory in a completely different direction. What if, instead of trying to simulate quote-unquote reality, we just simulate whatever the fuck could possibly be simulated from theorems and proofs of mathematics and physics?

That's another reason why it's ingenious, but it hasn't really been done yet. It takes a bit of an outsider who hasn't dove into all these concepts to the point that they think about it so rigidly and so anthropometrically, because the more and more you're talking to people about this stuff, the more and more you're reading literature about this stuff, the more and more you're getting people's approximations and speculations and this and that.

Which is why we want the framework to build along with like the architectures to be built along with deciding, you know, there's there's been v1s v2s v4s and none of these versions are actually contingent on the other, you know the next version of GOG after all the next round of battle testing, you know this framework against reality It might be something completely different. You know the goal isn't to just

Well, I think that's understood without me having to go on too much. And another thing, someone asked me where this idea came from. Well, I had the idea that, okay, if there was to be a unified theory of everything, how would it be proven? Would somebody just public a dissertation? You know, how would we know if we've really gotten the math theories that represent the universe? How would we know? Well, it's to just simulate a universe using those math theories, using those theorems.

So how would we know the standard model of physics works, right? Or the linear agent, how would we know if it works or not? Well, it's just simulate based off those models.

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